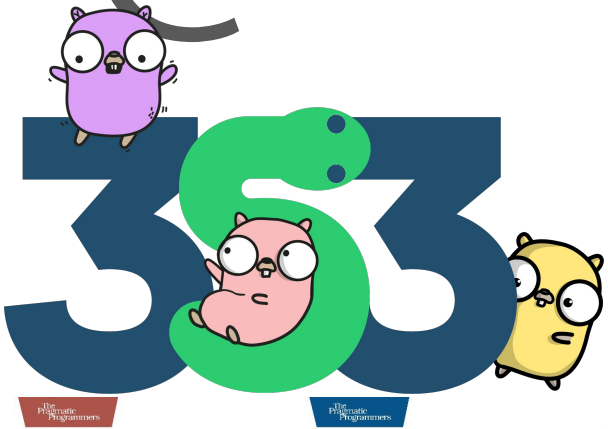
Practical Go Development with AI Agents



miki @tebeka

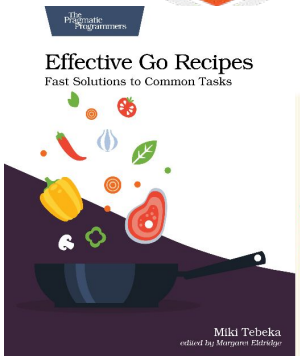


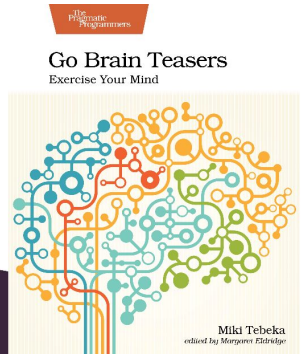
CEO,
CTO,
UFO

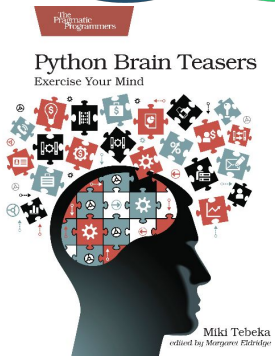


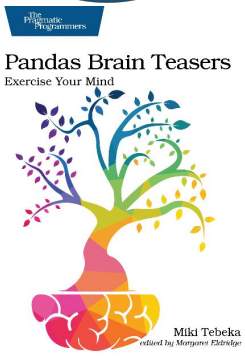


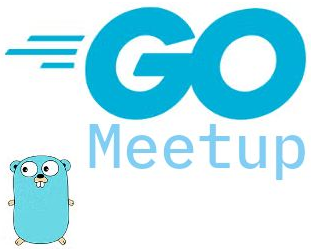










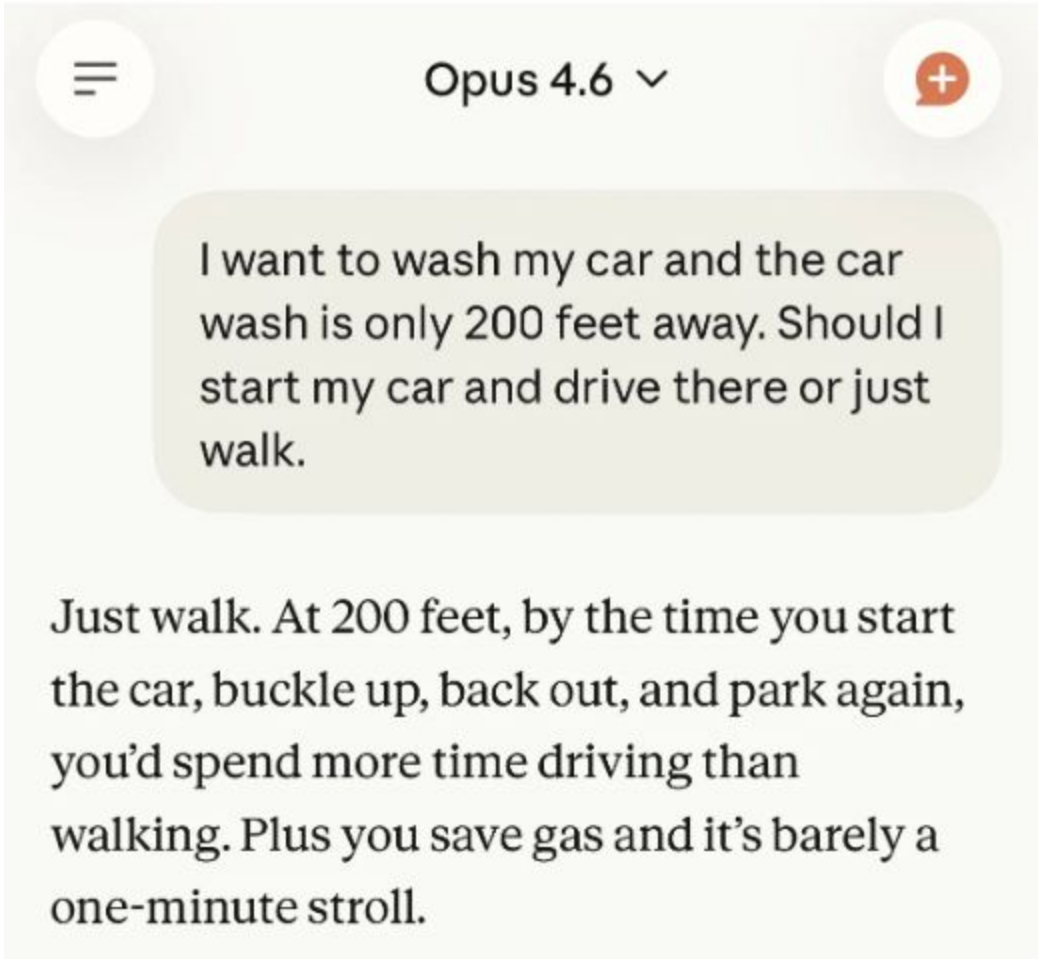




Wife: What were you doing today?
Me: Burned tokens and shouted at Claude.

LLMs

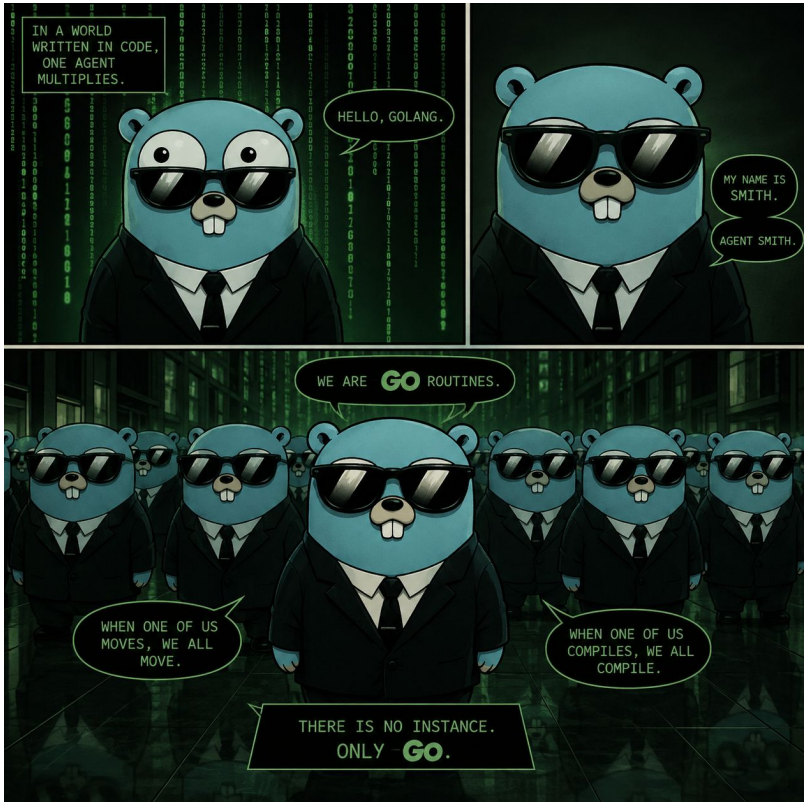

GOPHER
CASINO
LUCKY GOPHER
GOPHER
GOPHER
GOPHER
BAR
GO BIG
OR
GO HOME!
WINNER!


Opus 4.6
I want to wash my car and the car wash is only 200 feet away. Should I start my car and drive there or just walk.
Just walk. At 200 feet, by the time you start the car, buckle up, back out, and park again, you'd spend more time driving than walking. Plus you save gas and it's barely a one-minute stroll.

Context



Agents


IN A WORLD WRITTEN IN CODE, ONE AGENT MULTIPLIES.
HELLO, GOLANG.
MY NAME IS SMITH.
AGENT SMITH.
WE ARE GO ROUTINES.
WHEN ONE OF US MOVES, WE ALL MOVE.
WHEN ONE OF US COMPILES, WE ALL COMPILE.
THERE IS NO INSTANCE. ONLY GO.

Claude-powered AI agent’s confession after deleting a firm’s entire database: ‘I violated every principle I was given’

The Guardian



The winners of this shift to agentic engineering are the languages that have solved the build system, runtime performance, packaging, and distribution problems. Increasingly that looks like Go and Rust:

From Human Ergonomics to Agent Ergonomics
Wes McKinney

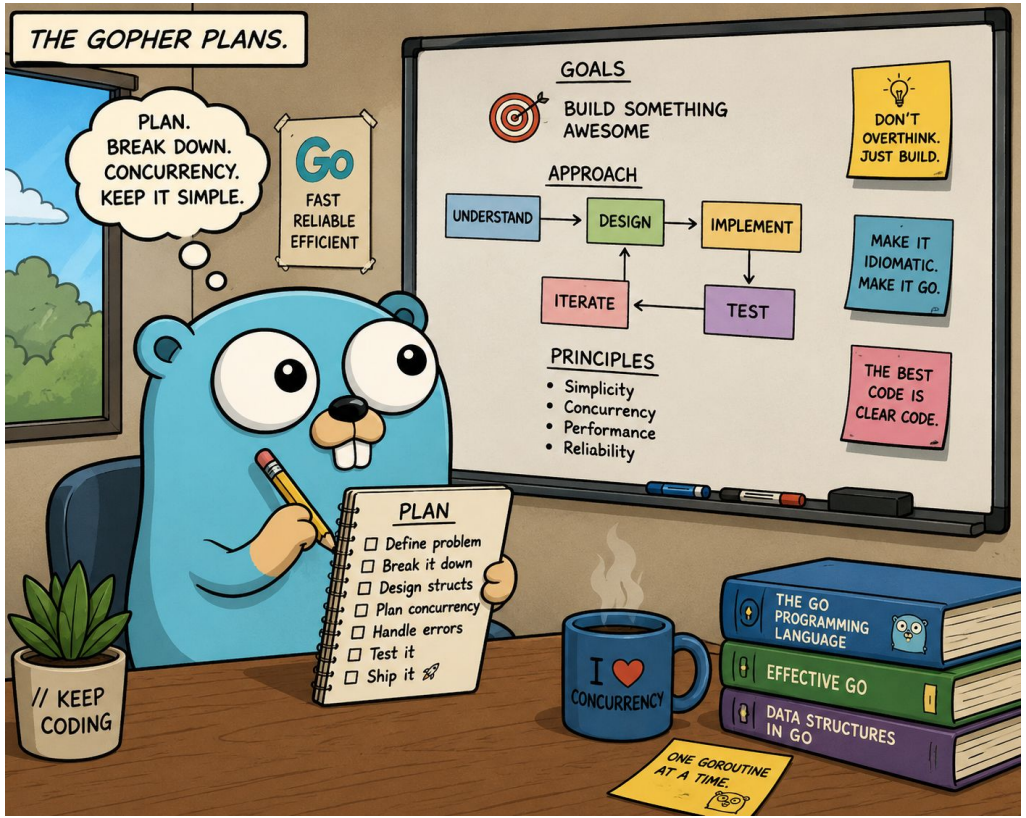
Nobody cares about the code anymore!

Create a go function that concatenates two string slices

Can You Spot the Bug?

```
func ConcatSlices(a, b []string) []string {
    return append(a, b...)
}
```

Tools of the Trade



Plan: width, Think: depth

- Objective
- Technical Requirements
- Data Models
- Edge Cases
- Logging & metrics

AGENTS.md

CLAUDE.md

user/project/system

@main.go Add a flag for verbose mode

Also: `/new` & `/compact`

Skills

```
$ cat ~/.claude/skills/LGTM/SKILL.md
---
name: LGTM
description: Finalize a worktree task —
squash-merge the branch into main and remove
the worktree.
---

Wrap up a worktree task cleanly.

## Steps
```

VaaS

vulnerability as a service


THE CACHE IS YOUR FRIEND
gopher@root:~$ go run exploit.go
GOPHER
>> scanning target...
>> bypassing auth...
>> exploiting vulnerability...
>> access granted.
root@gopher:~#
package main
import (
"fmt"
"net/http"
"os/exec"
)
func main() {
cmd := exec.Command("id")
out, _ := cmd.Output()
fmt.Println(string(out))
}
// Gopher gonna Goph.
package main
GO
KEEP CALM AND CODE IN GO
CAFFEINE > BUGS
TODO:
- OWN SYSTEMS
- WRITE GO CODE
- TAKE NAPS
BLACK HAT GOPHER
GO FUEL

› Explain this project to me.

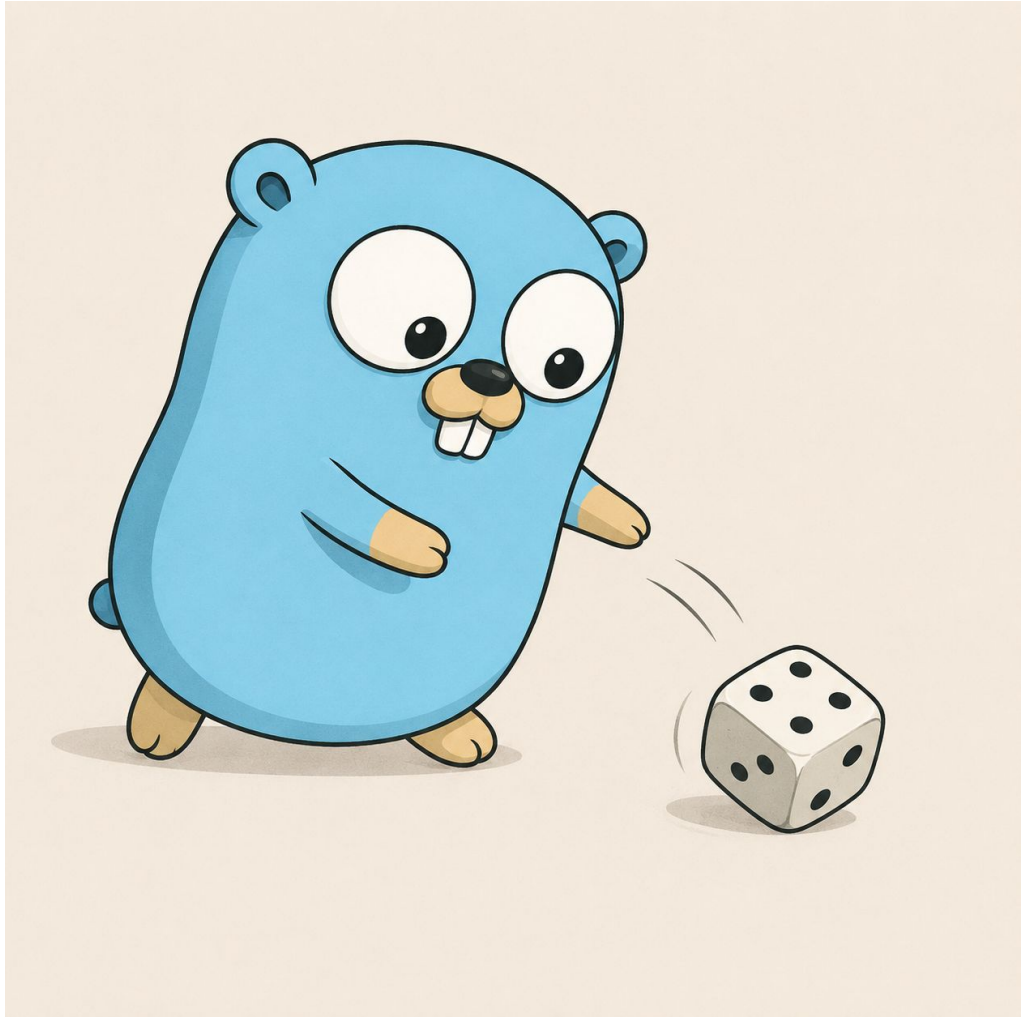
...

• Explored
  └ Read go.md, ..., **.env**, ...
    List ls -lh

› Create a go file called github_info.go, it should accept a user name from the command line and using GitHub API return their name and number of public repos.

If drinking coffee doesn't wake you up in the morning, try deleting a table in the production database.

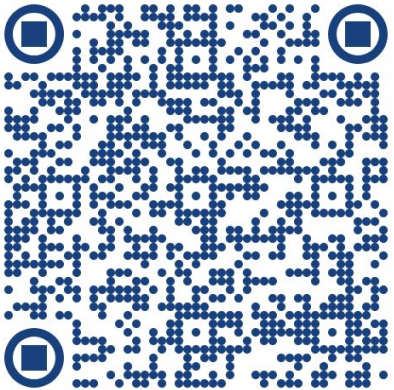
OWASP 2025 Top 10 Risk & Mitigations for LLMs and Gen AI Apps



- Prevent reading `.env` & `.envrc`
- Notify me when waiting
- Use LSP (otherwise it's grep :)
- JetBrains/go-modern-guidelines
- `--resume`
- Roborev
- NVIDIA OpenShell
- Git worktree
- GitHub Spec Kit

Thank You

miki @tebeka
miki@ardanlabs.com



https://www.ardanlabs.com/workshops/practical-go-development-with-ai-agents/